# D-Subsets Have Equal Coefficients: Filling in Rodina's Argument*

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## Section 4 — D-Subsets Have Equal Coefficients

The proof of $\mathrm{Tr}(\phi^3)$ uniqueness in [1, "$\mathrm{Tr}(\phi^3)$ subset zeros and uniqueness"] runs through D-subsets: collections of $n$-point diagrams obtained by attaching leg 1 in every valid way to a fixed $(n-1)$-point diagram. The argument has two key claims. The first — that each D-subset satisfies the 1-zero on its own (eqs. (20)–(21)) — is rigorous as stated. The second — that within each D-subset *all coefficients are equal* (eq. (22)) — is established only by example, illustrated for $k = 2$ at $n = 6$.

This section fills in the second claim for arbitrary $k$ and $n$. The argument is a single induction on adjacent pairs: for each pair of consecutive diagrams in a D-subset, two simultaneous limits isolate the pair and the master substitution $X_{2,j} = X_{1,j} - X_{1,3}$ from the 1-zero collapses the residual identity to $\alpha_j = \alpha_{j+1}$. Chaining gives the proposition.

### 4.1 Setup: D-subsets as fixed-$\beta$ slices of $B_i$

By the subset-vanishing theorem of §3, the BCFW homogeneity partition commutes with the zero condition: writing $B = \sum_i B_i$, each $B_i$ separately satisfies $B_i|_{Z_1} = 0$. Splitting variables as $x =$ (boundary props) and $y =$ (everything else), expand

$$B_i(x, y) \;=\; \sum_{\substack{\beta \\ |\beta|=(n-3)-i}} \; \sum_{\substack{\alpha \\ |\alpha|=i}} \frac{a_{\alpha\beta}}{x^\alpha\, y^\beta}.$$

Since $\ell(x) = 0$ does not depend on $y$, each fixed-$\beta$ piece must vanish on $Z_1$ on its own. *This fixed-$\beta$ piece is the D-subset:*

$$S_\beta \;=\; \sum_{\substack{\alpha \\ |\alpha|=i}} \frac{a_{\alpha\beta}}{x^\alpha\, y^\beta}, \qquad S_\beta\big|_{Z_1} \;=\; 0.$$

**Geometric reading.** The boundary propagators are the planar chords containing leg 1: chords of the form $X_{1,\ell}$ (separating leg 1 from one side of the diagram) and $X_{2,\ell}$ (separating legs 1 and 2 from the rest). A D-subset with $k$ boundary propagators contains $k + 1$ diagrams, one for each way to insert leg 1 into the corresponding $(n - 1)$-point skeleton. Writing the relevant indices in increasing order

$$3 \;=\; \ell_0 \;<\; \ell_1 \;<\; \cdots \;<\; \ell_{k-1} \;<\; \ell_k \;=\; n,$$

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with $\ell_1, \ldots, \ell_{k-1} \in \{4, \ldots, n-1\}$ arbitrary, the $i$-th diagram (for $i = 0, 1, \ldots, k$) has rational form

$$\frac{1}{X_{1,\ell_0}\, X_{1,\ell_1} \cdots X_{1,\ell_{i-1}} \cdot X_{2,\ell_{i+1}}\, X_{2,\ell_{i+2}} \cdots X_{2,\ell_k}},$$

with empty products understood as 1. (Note $X_{1,\ell_0} = X_{1,3}$ and $X_{2,\ell_k} = X_{2,n}$.)

**Proposition 1** (D-subset uniqueness)**.** *Let $S$ be a D-subset with $k \geq 1$ boundary propagators and coefficients $\alpha_0, \alpha_1, \ldots, \alpha_k$. If $S|_{Z_1} = 0$, then $\alpha_0 = \alpha_1 = \cdots = \alpha_k$.*

### 4.2 Worked cases

We illustrate the argument for $k = 1$ and $k = 2$ before stating it in general; both are special cases of the general induction step that follows.

**$k = 1$.** Two diagrams. Common denominator gives

$$S \;\propto\; \frac{\alpha_1}{X_{1,3}} + \frac{\alpha_2}{X_{2,n}} \;=\; \frac{\alpha_1\, X_{2,n} + \alpha_2\, X_{1,3}}{X_{1,3}\, X_{2,n}}.$$

On $Z_1$, the master substitution gives $X_{2,n} = X_{1,n} - X_{1,3} = -X_{1,3}$ (since $X_{1,n} = 0$ as an external leg). Substituting,

$$0 \;=\; \alpha_1 X_{2,n} + \alpha_2 X_{1,3} \;=\; (-\alpha_1 + \alpha_2)\, X_{1,3} \quad\Longrightarrow\quad \alpha_1 = \alpha_2.$$

**$k = 2$.** Three diagrams, parameterized by an internal index $\ell_1 \in \{4, \ldots, n-1\}$. Common denominator,

$$0 \;\propto\; \alpha_1\, X_{2,\ell_1} X_{2,n} \;+\; \alpha_2\, X_{1,3} X_{2,n} \;+\; \alpha_3\, X_{1,3} X_{1,\ell_1} \quad \text{on } Z_1. \tag{1}$$

On $Z_1$ the free variables are $X_{1,3}$ and $X_{1,\ell_1}$; the remaining boundary props are determined by $X_{2,\ell_1} = X_{1,\ell_1} - X_{1,3}$ and $X_{2,n} = -X_{1,3}$. Take two limits in turn:
*Limit* $X_{1,\ell_1} \to 0$. Then $X_{2,\ell_1} \to -X_{1,3}$ and the $\alpha_3$ term vanishes (it has $X_{1,\ell_1}$ as an explicit factor). What remains:

$$0 \;=\; \alpha_1(-X_{1,3})(-X_{1,3}) + \alpha_2 X_{1,3}(-X_{1,3}) \;=\; (\alpha_1 - \alpha_2)\, X_{1,3}^2 \;\Longrightarrow\; \alpha_1 = \alpha_2.$$

*Limit* $X_{2,\ell_1} \to 0$ *(i.e.* $X_{1,\ell_1} \to X_{1,3}$*).* Now the $\alpha_1$ term vanishes. What remains:

$$0 \;=\; \alpha_2 X_{1,3}(-X_{1,3}) + \alpha_3 X_{1,3} \cdot X_{1,3} \;=\; (-\alpha_2 + \alpha_3)\, X_{1,3}^2 \;\Longrightarrow\; \alpha_2 = \alpha_3.$$

This is exactly Rodina's eq. (22). The mechanism is: each limit kills all but two of the three terms, and the master substitution turns the remainder into a single linear relation $\propto (\alpha_a - \alpha_b)\, X_{1,3}^2$.

### 4.3 General $k$: the induction step

**Numerator after clearing.** Multiply $S$ through by the common denominator $\big(\prod_{a=0}^{k-1} X_{1,\ell_a}\big)\big(\prod_{b=1}^{k} X_{2,\ell_b}\big)$. The $i$-th term has its $X_{1,\ell_a}$ factors for $a = 0, \ldots, i-1$ in the denominator, so clearing leaves the *remaining* factors in the numerator. The numerator that must vanish on $Z_1$ is

$$\mathcal{N} \;=\; \sum_{i=0}^{k} \alpha_i \Big(\prod_{a=i}^{k-1} X_{1,\ell_a}\Big)\Big(\prod_{b=1}^{i} X_{2,\ell_b}\Big) \;=\; 0. \tag{2}$$

**Two simultaneous limits.** For each $j \in \{0, 1, \ldots, k-1\}$, take the joint limit

$$X_{1,\ell_{j+1}} \;\to\; 0, \qquad X_{2,\ell_j} \;\to\; 0.$$

On $Z_1$, $X_{2,\ell_j} = X_{1,\ell_j} - X_{1,3}$, so the second limit is achieved by $X_{1,\ell_j} \to X_{1,3}$, a free specialization. Both limits are legal.

Examine which terms of (2) vanish: the $i$-th term has first product $\prod_{a=i}^{k-1} X_{1,\ell_a}$ and second product $\prod_{b=1}^{i} X_{2,\ell_b}$.

- The first product contains the vanishing $X_{1,\ell_{j+1}}$ as a factor when $i \le j+1 \le k-1$, in which case the term picks up at least one factor of $X_{1,\ell_{j+1}} \to 0$.
- The second product contains the vanishing $X_{2,\ell_j}$ as a factor when $1 \le j \le i$, in which case the term picks up at least one factor of $X_{2,\ell_j} \to 0$.

For $i \le j-1$: only the first condition fires (one factor of $X_{1,\ell_{j+1}}$). The term is $O(X_{1,\ell_{j+1}})$ in the joint limit. For $i \ge j+2$: only the second condition fires (one factor of $X_{2,\ell_j}$). The term is $O(X_{2,\ell_j})$. For $i \in \{j, j+1\}$: *both* conditions fire (one factor of each). The term is $O(X_{1,\ell_{j+1}} X_{2,\ell_j})$.

So the leading-order content of (2) in the joint limit at the rate $O(X_{1,\ell_{j+1}})$ is

$$\sum_{i \le j-1} \alpha_i \Big(\prod_{a=i}^{k-1} X_{1,\ell_a}\Big)\Big(\prod_{b=1}^{i} X_{2,\ell_b}\Big) \;=\; 0, \tag{3}$$

and at the rate $O(X_{2,\ell_j})$ is

$$\sum_{i \ge j+2} \alpha_i \Big(\prod_{a=i}^{k-1} X_{1,\ell_a}\Big)\Big(\prod_{b=1}^{i} X_{2,\ell_b}\Big) \;=\; 0. \tag{4}$$

These are independent identities (corresponding to different orders of vanishing) and they are inductively redundant for the pairs $j-1$ and $j+1$. The new content of the joint limit $j$ is the cross-term at rate $O(X_{1,\ell_{j+1}} X_{2,\ell_j})$, which is the sum of the $i = j$ and $i = j+1$ terms:

$$\alpha_j \Big(\prod_{a=j}^{k-1} X_{1,\ell_a}\Big)\Big(\prod_{b=1}^{j} X_{2,\ell_b}\Big) \;+\; \alpha_{j+1}\Big(\prod_{a=j+1}^{k-1} X_{1,\ell_a}\Big)\Big(\prod_{b=1}^{j+1} X_{2,\ell_b}\Big) \;=\; 0. \tag{5}$$

**Reducing the cross-term.** Pull out the common factor $\big(\prod_{a=j+1}^{k-1} X_{1,\ell_a}\big)\big(\prod_{b=1}^{j} X_{2,\ell_b}\big)$, leaving

$$\alpha_j\, X_{1,\ell_j} \;+\; \alpha_{j+1}\, X_{2,\ell_{j+1}} \;=\; 0.$$

Under the limit, $X_{1,\ell_j} \to X_{1,3}$ (from $X_{2,\ell_j} \to 0$) and $X_{2,\ell_{j+1}} = X_{1,\ell_{j+1}} - X_{1,3} \to -X_{1,3}$ (from $X_{1,\ell_{j+1}} \to 0$). Substituting,

$$\alpha_j\, X_{1,3} \;+\; \alpha_{j+1}\,(-X_{1,3}) \;=\; (\alpha_j - \alpha_{j+1})\, X_{1,3} \;=\; 0 \implies \alpha_j = \alpha_{j+1}.$$

**Boundary cases $j = 0$ and $j = k-1$.** The argument above assumed $1 \le j \le k-2$ so that both vanishing variables $X_{1,\ell_{j+1}}$ and $X_{2,\ell_j}$ are interior. For $j = 0$, take $X_{1,\ell_1} \to 0$ alone (the other "limit" $X_{2,\ell_0} \to 0$ is automatic since $\ell_0 = 3$ and $X_{2,3}$ is degenerate). The cross-term reduces to $\alpha_0\, X_{1,\ell_0} + \alpha_1\, X_{2,\ell_1} = 0$, which under the limit gives $\alpha_0 X_{1,3} - \alpha_1 X_{1,3} = 0$, hence $\alpha_0 = \alpha_1$. Symmetrically for $j = k-1$, take $X_{2,\ell_{k-1}} \to 0$ alone, giving $\alpha_{k-1} = \alpha_k$.

**Chaining.** Applying the result to each $j \in \{0, 1, \ldots, k-1\}$:

$$\alpha_0 \;=\; \alpha_1 \;=\; \alpha_2 \;=\; \cdots \;=\; \alpha_k. \qquad \square$$

### 4.4 What this fills in vs. Rodina

Rodina's eq. (22) gives the $k = 2, n = 6$ case explicitly and asserts "the whole process can be carried out for each D-subset." The induction above makes that assertion precise. The mechanism is uniform across $k$: for each adjacent pair $(\alpha_j, \alpha_{j+1})$ in any D-subset of any size, the simultaneous limit $X_{1,\ell_{j+1}}, X_{2,\ell_j} \to 0$ on $Z_1$ isolates exactly those two coefficients in the cross-term (5), and the master substitution collapses the residue to $(\alpha_j - \alpha_{j+1})X_{1,3} = 0$.

The argument never invokes residue cuts at non-boundary propagators (Rodina's eq. (21)); those cuts are an independent argument, used to establish that each D-subset satisfies the zero on its own. Here we are proving uniqueness within an already-vanishing D-subset, and the limit-on-boundary-chords mechanism above is what does the work.

Combined with Rodina's graph-connectivity argument (a leg-removal induction on $n$, rigorous as stated in his "Amplitude uniqueness" paragraph), Proposition 1 closes the uniqueness theorem for $\mathrm{Tr}(\phi^3)$.

## References


[1] L. Rodina, "Hidden zeros are equivalent to enhanced ultraviolet scaling and lead to unique amplitudes in $\mathrm{Tr}(\phi^3)$ theory," arXiv:2406.04234.